SOFTWARE ENGINEERING LAB 3

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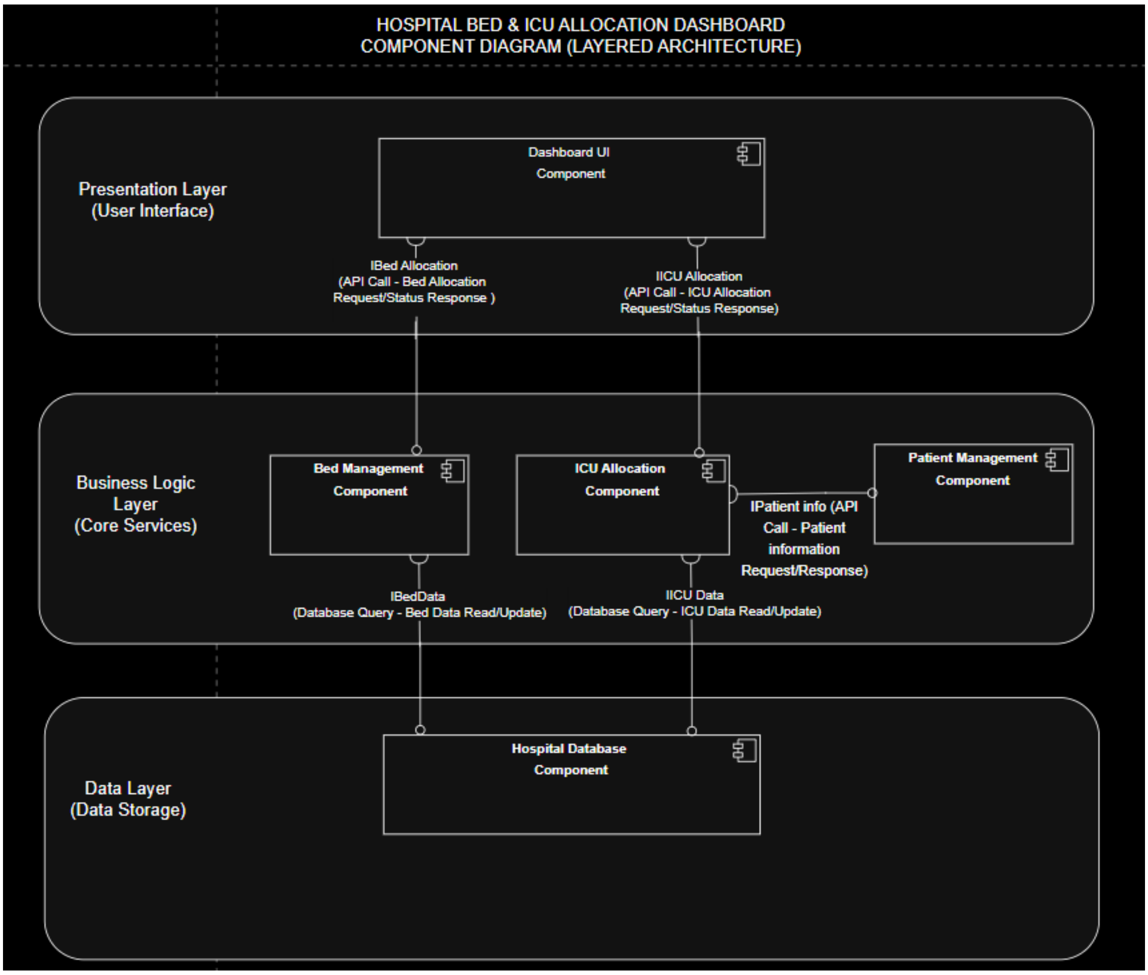
SECTION : E

ARCHITECTURE SELECTION & JUSTIFICATION :

Architecture Selection: I chose Layered Architecture for the Hospital Bed & ICU Allocation Dashboard.

CRITERIA	JUSTIFICATION
Architecture Selection	I chose Layered Architecture for the Hospital Bed & ICU Allocation Dashboard. The system is divided into three layers: Presentation Layer, Business Logic Layer, and Data Layer**.**
Architectural Choice	The Presentation Layer contains the Dashboard UI Component for hospital staff interaction. The Business Logic Layer contains the Bed Management, ICU Allocation, and Patient Management Components. The Data Layer contains the Hospital Database Component for storing and retrieving hospital information.
Reason 1 – Clear Separation of Responsibilities	Layered Architecture clearly separates the user interface, hospital allocation logic, and data storage. For example, the Dashboard UI displays bed and ICU information, while the Bed Management and ICU Allocation Components handle the actual allocation decisions. This makes the system easier to understand and organize.
Reason 2 – Easier Maintenance	Different parts of the hospital system can be maintained or modified independently. For example, changes to the Dashboard UI can be made without directly changing the Hospital Database or the bed and ICU allocation logic.
Security Advantage	The Dashboard UI does not directly access sensitive patient and hospital data. Requests first go through the Business Logic Layer before accessing the Hospital Database. This provides better control over access to sensitive patient and allocation information.
Performance Benefit	The separation of responsibilities allows the Dashboard UI, business components, and database to focus on their specific tasks. Bed and ICU allocation requests are processed by the Business Logic Layer, while the Data Layer handles data storage and retrieval, helping the system operate efficiently

COMPONENT DIAGRAM :



Components Used (5) :

1.Dashboard UI Component:
Provides the user interface for hospital staff to view the availability of hospital beds and ICU beds. It also allows staff to send bed and ICU allocation requests to the system.

2.Bed Management Component:
Manages normal hospital bed availability and allocation. It checks available beds and updates the bed status after allocation or release.

3.ICU Allocation Component:
Manages ICU bed availability and handles ICU allocation requests. It checks ICU availability and updates the ICU status after allocation or release.

4.Patient Management Component:
Manages patient-related information required during the allocation process. It retrieves and provides patient details to other components when needed.

5.Hospital Database Component:
Stores information related to hospital beds, ICU beds, patients, and allocation records. It also provides data to the Bed Management and ICU Allocation components.

Interfaces (5) :

1.IBedAllocation:
Allows the Dashboard UI Component to communicate with the Bed Management Component. It is used to request bed availability, bed allocation, and receive allocation status.

2.IICUAllocation:
Allows the Dashboard UI Component to communicate with the ICU Allocation Component. It is used to request ICU availability and ICU bed allocation services.

3.IPatientInfo:
Allows the ICU Allocation Component to obtain required patient information from the Patient Management Component. It supports retrieving patient details needed during allocation.

4.IBedData:
Allows the Bed Management Component to access bed-related information from the Hospital Database. It is used to read bed data and update the status of beds.

5.IICUData:
Allows the ICU Allocation Component to access ICU-related information from the Hospital Database. It is used to retrieve ICU data and update ICU availability status.